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Purpose/Policy:

The purpose of this standard operating procedure is to understand how the Palletizer works, as well as operation, changeover, cleaning, and troubleshooting.

Scope:

The palletizer is comprised of several different sections. Once section is where you load empty pallets into the palletizer. Cases are fed into the palletizer via conveyors which are then arranged to build layers of cases. Once a layer of cases is built, they are dropped onto an empty pallet on a turn table that has rollers. Once a full pallet is built the pallet is rolled out on chains through a light curtain and is ready to be removed from the palletizer.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- E-Stop – Emergency Stop button
- Palletizer – Machine that stacks finished cases onto empty pallets at the end of the can line

Authorities/Responsibilities:

The primary role of the palletizer operator is to load wooden pallets into the palletizer as well as keeping an eye on cases as they enter the palletizer and are stacked. The operation of the palletizer can be done by an entry level packaging operator after proper training from a senior member of the can line.

Resources and Materials:

Operations:

- Wooden pallets
- Plastic wrap
- Staple gun
- Cardboard sheets

Basic Cleaning:

- Damp cloth
- Broom and dustpan

Deep Clean:

- Refer to SOP Document # CL###
    - Link to document goes here.

Troubleshooting:

- See maintenance in the event of a stuck pallet.

PPE Required:

- No formal PPE required, suggested nitrate gloves for cleaning.

Procedures:

Setting up the Palletizer:

1. Make sure the empty pallet dispenser has at least 3 empty pallets loaded into it.
2. Check to make sure that the plastic film dispenser has a roll of plastic film loaded up and threaded through the dispenser correctly.
3. Next make sure all E-Stops are pulled out.
4. Once these things are set up press the blue reset button. This will kick on the air supply for the palletizer.
5. Press and hold the green start button until the alarm stops. This will bring in an empty pallet and dispense some plastic wrap onto the pallet. It will also start up the infeed conveyors.

Operation of the Palletizer:

1. Once the palletizer is running, to keep it running you must load in wooden pallets.
2. Occasionally the palletizer will run out of shrink wrap and need to be reloaded.
    2.1 In order to change out the shrink wrap you will need to press the red stop button.
    2.2 Once the palletizer is stopped go over to the side door where the shrink wrap dispenser is located and open the doors.
    2.3 Remove the cardboard roll of the empty roll.
    2.4 Replace with a new roll making sure orientation matches the diagram on the shrink wrap dispenser.
    2.5 Open up the shrink wrap dispenser by using the small black lever above the rollers but under the yellow top of the shrink wrap dispenser.
    2.6 Now that the shrink wrap has been replaced close the doors and go back to the HMI and press the blue reset button and then hold the green start button once it is reset.
3. After a pallet is finished being built and wrapped it will be ejected out via rollers and then moved along the chain conveyor. The pallet will then need to be removed with a forklift.

Changeover Guide:

1. On the HMI press “Current Pattern”.
2. Use the up and down arrows to select the pattern needed for the corresponding run.
3. Press “↲”.

4. Press the button to the right of the “use this pattern in the machine” ▲.
5. Wait for the screen to say, “New pattern loaded, Pres Start and Stop at the same time”.
6. Press the start and stop button at the same time to load the pattern.
7. Once the pattern is loaded you may press the green start button to start the machine.

Shutting Down Palletizer:

1. Ensure all cases are cleared out.
2. Press the stop button on the side of the HMI.
3. Press the E-Stop.

Basic Cleaning of the Palletizer:

1. When the machine is stopped it can be cleaned using a damp cloth to wipe off any beer that might have sprayed on any photo-eyes.
2. With a dustpan and broom sweep up any bits of wood that might be left behind from pallets.

Deep Cleaning:

Outline the deep cleaning process for the SOP equipment. See below for example:

1. Refer to SOP Document # CL###
    1.1 Link to document goes here.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

1. In the event of a broken shrink wrap fault there are a couple of possibilities.
    1.1 If the shrink wrap is actually broken, stop the machine and open up the side door.
    1.2 Open up the shrink wrap dispenser and pull enough extra length to tuck under a case on the pallet.
    1.3 Close up the shrink wrap dispenser and restart the palletizer.
    1.4 It also possible that the shrink wrap has gotten caught up on one of the shrink wrap dispenser rollers.
        1.4.1 In this event it is important to be careful cutting off the shrink wrap and not damage the roller.

2. If when the palletizer is stopped a case is in the transition area it will ask that the case be removed.
   - 2.1 Open the doors and remove the case.
   - 2.2 Once the case is removed close the doors and go over to the HMI.
   - 2.3 On the HMI press the “Reset case Count”.
   - 2.4 Then press the blue reset button and then press the green start button.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
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002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

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